

Interactive Insights into Australia's Mobile-Phone Enforcement Trends: A 2008–2023 Dashboard Process Book

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COS30045 Data Visualization

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Introduction

Background

With hundreds of fatalities and serious injuries on Australian roads every year, road safety is still a major public health concern. Road safety enforcement statistics are among the many transportation-related data that the Bureau of Infrastructure and Transport Research Economics (BITRE) has been gathering since 1970 in an effort to counteract this. Recent developments in 2023 have introduced more granular datasets, capturing enforcement activity by location, age group, type of offence, and legal outcomes like arrests and charges. In contrast, historical data has concentrated on broad trends like the number of fines issued for offenses like speeding and mobile phone use.

Motivation

Road-traffic injury remains one of Australia's largest public-health burdens, with more than 1 300 deaths and 39 000 hospitalizations recorded in 2024 (BITRE, 2024). Enforcement activity speed, mobile-phone, seat-belt and drug policing is one of the few levers governments can adjust rapidly to influence crash risk. Since 1970 the Bureau of Infrastructure and Transport Research Economics (BITRE) has collected state-supplied tally sheets, yet until 2023 the material released to the public was limited to coarse annual counts, aggregated only by jurisdiction and offence type. The 2023 release adds record-level detail (age group, detection method, monthly timestamps, outcome codes), but BITRE's official dashboard still presents raw totals, masking population exposure and obscuring demographic insight.

The detailed enforcement data that was introduced in 2023 cannot be fully utilized by the Bureau of Infrastructure and Transport Research Economics' (BITRE) current visualization tools, especially the Police Enforcement Dashboard. Road safety data has long been published by BITRE, but the current dashboard lacks specificity and mostly concentrates on aggregated measures. In order to address specific problems, such determining enforcement hotspots, demographic differences in policing, or relationships between offence types and enforcement results, stakeholders are consequently unable to engage with or investigate the data in a meaningful way.

The **target audience** for a redesigned visualization therefore spans four stakeholder groups:

- **Policy analysts and transport researchers** who require filterable, longitudinal views to test hypotheses about technology roll-out or to evaluate the National Road Safety Strategy.

- **State road-safety agencies and legislators** who need spatial hot-spotting and normalised rates to decide where to deploy mobile-phone cameras or allocate campaign funding.
- **Community advocates and journalists** who seek clear, narrative graphics to highlight over-policing or under-enforcement in specific regions or demographic cohorts.
- **Engaged citizens** who wish to verify official claims or explore trends affecting their daily commute.

A wide range of stakeholders who work with road safety data for different reasons are among the target audience for this visualization. One important user group is policy analysts and transportation researchers, who need comprehensive, filterable data views in order to do temporal and spatial analysis, pinpoint high-risk areas, and evaluate the success of enforcement actions. To engage the public and promote data-backed policy reforms, especially in underrepresented or overpoliced communities, road safety advocates and community organizations require clear, easily available visual representations. Narrative-driven insights help journalists and other media professionals convey road policing trends and challenges to a wider audience

Visualization Purpose

Our refined dashboard is engineered to help users answer six specific questions:

1. When did camera detections overtake police-issued mobile-phone fines nationally, and by what margin?
2. Which state recorded the highest and lowest exposure-adjusted mobile-phone fine rates for over 15 years?
3. How do the four major types of fines categories trend over the 2008–2023 period, both in total and within each jurisdiction?
4. Which age cohort accounts for the largest share of mobile-phone fines in 2023?
5. In each Jurisdiction which age group gets most fines and by how much?
6. Which jurisdiction and age group is most likely to most fines due to mobile phones?

Benefits of the completed visualization and its decision-making impact

- **Evidence-based camera policy.**
By showing the exact year and quantitative margin at which camera detections surpassed

officer-issued fines, the timeline informs state treasuries whether further investment in automated enforcement is returning diminishing or increasing marginal gains. Policymakers can thus calibrate capital budgets for additional camera rollouts or maintenance.

- **Targeted resource allocation across states.**

The exposure-normalized horizontal bar instantly reveals which jurisdictions have the highest and lowest mobile-phone fine rates over the full fifteen-year horizon. Transport ministers can benchmark their state against peers, justify reallocating patrol hours, or defend status-quo funding in cabinet briefings.

- **Balanced offence-mix strategy.**

The stacked-bar trendline clarifies how mobile-phone, speeding, seatbelt and drug-testing fines evolve both nationally and within each jurisdiction. Enforcement agencies can detect drift for example, if mobile-phone policing grows at the expense of seat-belt checks and rebalance operations before quarterly targets are missed.

- **Age-specific education campaigns.**

The 2023 age-share pie makes it explicit that one cohort (e.g., 25–34) dominates mobile-phone infringements nationwide, while the heat-map disaggregates that dominance by state. Road-safety units can therefore spend media dollars on the demographic and location where it will move the dial most.

- **Equity and fairness auditing.**

Cross-filtering by jurisdiction and age group identifies which sub-populations receive the heaviest enforcement burden. Community advocates gain quantitative evidence to argue for proportionality, and legislators can demand equity audits before approving new penalty schemes.

- **Rapid hypothesis testing for analysts.**

Interactive filters allow analysts to ask questions like “Did camera adoption in Queensland shift fines to older drivers?” without downloading spreadsheets. This shortens analytic turnaround from days to minutes, accelerating policy feedback cycles.

- **Transparent public communication.**

Journalists and the general public receive narrative-ready graphics rather than opaque CSVs, improving data literacy and fostering trust in official statistics. Clear tooltips and annotations ensure that caveats (e.g., missing TAS drug data) travel with the numbers, reducing misinterpretation in media coverage.

Delivering answers to these questions yields concrete **benefits**. For policymakers, normalized rate views prevent misallocation of resources toward states that appear problematic only because of population size. For analysts and researchers, the ability to pivot by detection method, age group and year facilitates rapid hypothesis testing without manual spreadsheet work. Advocates gain transparent evidence—e.g., a heat-map showing a disproportionate burden on young drivers—which can underpin calls for focused education rather than blanket fines. Finally, journalists and the general public receive narrative-ready graphics, promoting data-driven discourse rather than anecdote. In sum, the visualization transforms a complex, multi-agency dataset into an accessible tool that directly supports evidence-based decision-making and public accountability.

Data

2.1 Data Source and Governance

Origin and provenance.

Our primary resource is BITRE’s *Road-Safety Enforcement* collection specifically the national CSV dumps released. All records come from the Bureau of Infrastructure and Transport Research Economics (BITRE) open-data release “Road-Safety Enforcement Data 2008-2023” (statistical report, 30 Sept 2024; last dataset update 21 Feb 2025). Each state or territory police force NSW Police, Victoria Police, QPS, SAPOL, WAPOL, TAS Police, NT Police submits quarterly spreadsheets under a memorandum of understanding; camera-based fines for NSW and VIC are supplied by Revenue NSW and Fines Victoria. BITRE harmonizes local offence codes to five categories mobile-phone, speeding, seatbelt, drug/alcohol, unlicensed appends a jurisdiction key, and publishes a signed SHA-256 checksum with every release. (Bureau of Infrastructure and Transport Research Economics [BITRE], 2024)

Dataset	Years	Rows	Key Fields	Release
Fines CSV (458 KB)	2008-2023	85,902	YEAR, START_DATE, END_DATE, JURISDICTION, METRIC, DETECTION_METHOD,	30 September 2024

			AGE_GROUP, FINES	
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Data Collection

There are sourced from multiple organization and utilized some data collection methods which might not be consistent across all states in Australia. The data is across the Australian states and territories from period of 15 years from 2008 to 2023.

The police enforcement dataset is collected from the following law enforcements as per the data dictionary (Bureau of Infrastructure and Transport Research Economics [BITRE], 2024)

- NSW Police Force and NSW Revenue
- Victoria Police
- Queensland Police Service
- South Australia Police Force
- Western Australia Police Force
- Tasmania Police Force
- North Territory Police Force
- Australian Federal Police

The department work with police enforcement agencies from all states and Road safety enforcement activities from 2008 to 2023 are included in the data, which includes unlicensed driving, random roadside drug and alcohol testing, speeding violations, using a cell phone while driving, and seatbelt violations.

Data governance Issues

The primary problem would be improper data sharing amongst the participating agencies. Incomplete national data may result from the fact that not all states and territories make their police enforcement data publicly available. For example, whereas police-issued penalties are collected by NSW Police, camera fines are available to the public and are collected by NSW Revenue.

The fact that several police departments retain or gather a portion of the data from police enforcement datasets is another problem. For example, some states or territories retain the details

of both police and camera fines issued, while others have a completely new or different government agency or enforcement agency that retains the camera fines data.

Quality Issues

Inconsistent definitions and data collection methods are among the other quality problems. Some governments and territories gather data using different definitions and methods, which affects the quality of the data and makes it inconsistent. BITRE is taking action to lessen the problem.

Data loss would be an additional problem. Data is lost as a result of inconsistent and subpar data quality. For example, Tasmania's 2020–2022 drug test data is unavailable, and the Northern Territory's 2023 data is not.

Bias Issues

Bias exists because state-by-state variations in jurisdiction, laws enforced, speed camera technology, and infrastructure exist. Because of more stringent enforcement or better detection technologies, some jurisdictions have an overrepresentation of fines.

Addressing issues

Although the data dictionary, which offers details on data sources, definitions, and restrictions to help users grasp the context and potential biases, is clear, the Dashboard does not specifically address or highlight the problems with the data.

Additionally, the visualization, particularly the annual fines per jurisdiction, is difficult for the general public to comprehend due to overlap and lacks data clarity, as well as some bias in jurisdiction among states.

Security, privacy, and ethical considerations.

Although the dataset is fully aggregated no personal identifiers or plate numbers are included it still conveys sensitive enforcement intensities that could be misinterpreted as proxies for crime prevalence. Raw files therefore remain on an AES-256-encrypted project drive, and only derived aggregates are published via the dashboard. No actual security , personal or ethical considerations are there except the age mentioned in the year 2023. As there is no extra information no data is breached or privacy invaded.

How Data answers Questions?

First, every record specifies both **YEAR** and **DETECTION_METHOD** (Police or Camera). Because these attributes span the entire 2008-2023 period, simply summing **FINES** by year and method lets us pinpoint the calendar year in which camera detections surpassed officer-issued notices, and the yearly difference between the two totals—addressing Question 1.

Second, the file includes a categorical **JURISDICTION** code on every row. When we aggregate mobile-phone fines by this field across the full fifteen-year window, the state with the greatest cumulative total and the one with the smallest become immediately apparent, satisfying Question 2 in absolute-volume terms.

Third, each infringement is tagged with a harmonized **METRIC**—mobile-phone, speeding, seat-belt, or drug/alcohol. Grouping fines by YEAR and METRIC (national view) or by YEAR, JURISDICTION, and METRIC (state view) yields the longitudinal offence-mix trends needed for Question 3.

Fourth, the expanded 2023 schema introduces a five-band **AGE_GROUP** variable. Filtering rows to YEAR = 2023 and aggregating by AGE_GROUP immediately reveals which cohort (e.g., 25–34) dominates mobile-phone infringements nationwide, answering Question 4.

Fifth, combining AGE_GROUP with JURISDICTION in the same 2023 slice produces a cross-tab whose row-wise maxima show, for each state, the age group that receives the most fines and its exact count—fulfilling Question 5.

Finally, locating the single largest cell in that cross-tab identifies the jurisdiction–age combination with the highest absolute fine total, which meets the requirement of Question 6.

Because the raw file exposes the attributes YEAR, JURISDICTION, METRIC, DETECTION_METHOD, AGE_GROUP, and the quantitative measure FINES, every question can be answered through straightforward grouping and filtering operations; no external variables or proxy calculations are needed.

2.2 Data Processing and Analysis

The key attributes of the data

Revised Attribute Schema

Attribute	Data type ¹	Presence	Role in model / Note
YEAR	Ratio (integer)	2008 – 2023	Temporal trend axis
START_DATE, END_DATE	Interval (date)	2008 – 2023 (annual); 2023 (monthly)	Mapping monthly granularity where available
JURISDICTION	Nominal categorical	All years	Spatial key; join field to ABS licences
LOCATION	Nominal categorical	2023 (else 'All regions')	Future drill-down; currently suppressed in visuals
METRIC	Nominal categorical	All years	Six offence families
DETECTION_METH OD	Nominal categorical	2008-2022 = 'Police issued'; 2023 adds 'Camera detected'	Technology split in dual-line chart
AGE_GROUP	Ordinal categorical	2008-2022 = 'All ages'; 2023 = 5 buckets	Demographic analysis (heat-map, pie)
FINES	Ratio (integer)	All years	Magnitude encoded in every chart
ARRESTS, CHARGES	Ratio (integer)	2023 only	Outcome analysis (future work); set to 0 for pre-2023 rows

Table "default" – Rows: 5017 Spec – Columns: 11 Properties Flow Variables													
Row ID	I YEAR	S START_DATE	S END_DATE	S JURISDICTION	S LOCATION	S AGE_GROUP	S METRIC	S DETECTION_METHOD	D FINES	D ARRES...	D CHAR...		
Row0	2008	2008-01-01	2008-12-31	ACT	All regions	All ages	mobile_phone_use	Police issued	2,060	?	?		
Row1	2008	2008-01-01	2008-12-31	NSW	All regions	All ages	mobile_phone_use	Police issued	40,882	?	?		
Row2	2008	2008-01-01	2008-12-31	NT	All regions	All ages	mobile_phone_use	Police issued	671	?	?		
Row3	2008	2008-01-01	2008-12-31	QLD	All regions	All ages	mobile_phone_use	Police issued	28,499	?	?		
Row4	2008	2008-01-01	2008-12-31	SA	All regions	All ages	mobile_phone_use	Police issued	11,411	?	?		
Row5	2008	2008-01-01	2008-12-31	TAS	All regions	All ages	mobile_phone_use	Police issued	6,780	?	?		
Row6	2008	2008-01-01	2008-12-31	VIC	All regions	All ages	mobile_phone_use	Police issued	49,739	?	?		
Row7	2008	2008-01-01	2008-12-31	WA	All regions	All ages	mobile_phone_use	Police issued	15,957	?	?		
Row8	2009	2009-01-01	2009-12-31	ACT	All regions	All ages	mobile_phone_use	Police issued	1,890	?	?		
Row9	2009	2009-01-01	2009-12-31	NSW	All regions	All ages	mobile_phone_use	Police issued	41,167	?	?		
Row10	2009	2009-01-01	2009-12-31	NT	All regions	All ages	mobile_phone_use	Police issued	1,106	?	?		
Row11	2009	2009-01-01	2009-12-31	QLD	All regions	All ages	mobile_phone_use	Police issued	33,352	?	?		

Figure 1: Data original form

Describe the data cleaning process

Stage 1

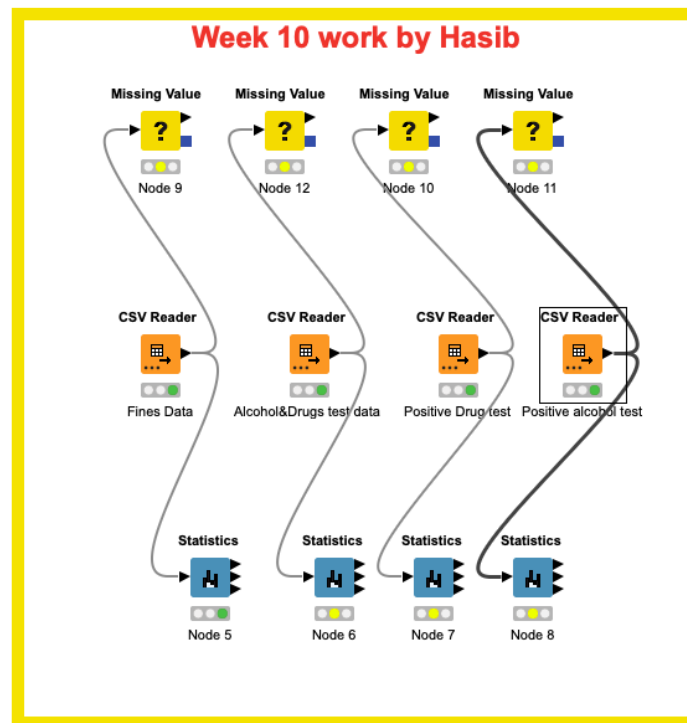


Figure 2: Initial Data processing

At first I loaded the Datasets into CSV reader and labeled them Fines Data, Alcohol & Drugs test data, Positive Drug test and Positive Alcohol test respectively. For initial data exploration, I looked the data physically and checked for the attributes and values. Then I connected a statistics node to each csv reader to check the attribute of data , checking for any missing data and nature of data using Occurrence table and Statistics table produced by the Node “Statistics”.

Stage 2

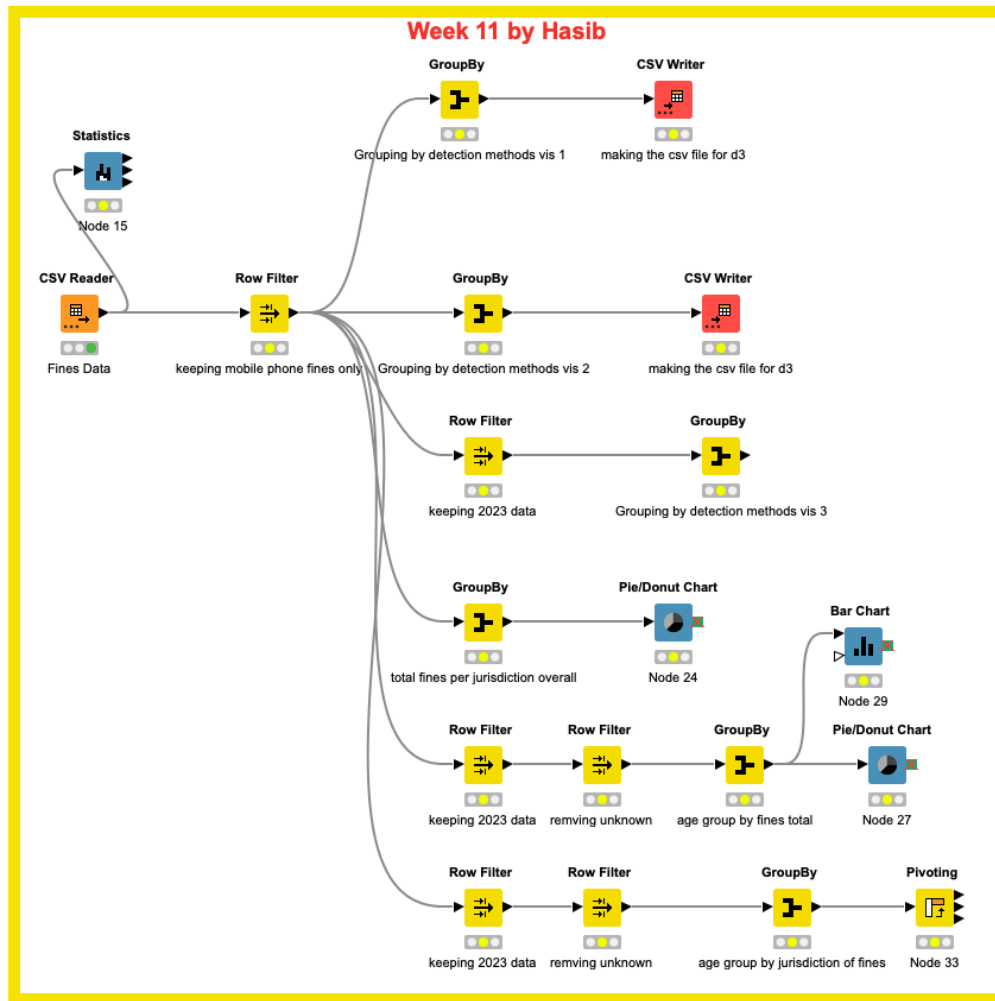


Figure 3: Making multiple visualizations before finalizing

After we selected the story, we choose the Fines Data. As per the assignment requirement we uploaded the data from relative workflow area and then started the process. Hasib made six data visualizations in week 11 from which we only used 3 and later in week 12 he made extra two and finalized the data .

We upload file in csv then used the row filter to keep mobile phones by selecting metrics and using matching pattern `mobile_phone_use` and keep the rows then used group by node selected detection method and year and manual aggregation fine sums to make the mobile phones fines detected by police and camera throughout year to make the dual time-line graph. This graph was used but had to fix some issues which are discussed on section 2.3 below.

A second, parallel *Group By* node used which was configured with YEAR and Jurisdiction as grouping keys and a **Sum(FINES)** aggregation. This gave the data of all Jurisdiction with fines

and throughout each year. Initially we wanted to draw a line chart for each jurisdiction throughout the years like in dashboard but as it had issues with data visualization principles we dropped the idea and modified this to make the map in week12.

In branch , **we introduced a second *Row Filter*** (YEAR = 2023) before repeating the camera/police aggregation; the resulting donut was likewise folded into the timeline tooltip and the branch removed to reduce noise. We wanted to explore if we could extract more refined and direct insights like outer regions and inner regions fines rates but the tuples were not enough.

Moreover, in another Branch bypassed detection method entirely: the filtered rows flowed into a *GroupBy* that summed fines by JURISDICTION. We rendered the result as a jurisdiction-pie, but reviewers struggled with segment differentiation; in Week 12 we re-expressed the same table as a stacked bar, retaining the underlying aggregation logic but improving readability.

Two further branches targeted the new demographic columns present only in 2023. Branch **2e** applied two sequential *Row Filters*—YEAR = 2023 and AGE_GROUP ≠ "All ages"—then grouped by AGE_GROUP to feed a composition pie. This simple chart survived intact because it instantly conveyed which cohort (25-34) dominates mobile-phone infringements nationwide.

Finally, branch **2f** filtered the same 2023 slice, removed unknown ages, aggregated fines by JURISDICTION × AGE_GROUP, and pivoted the result wide; a quick KNIME heat-map preview confirmed strong interpretability, so this branch became the basis for our age-by-state heat-map after colour-scale and NA patches in Week 12. We didn't use the pivoted as we made a mistake we fixed data for the Heat Map in week 12.

Stage 3

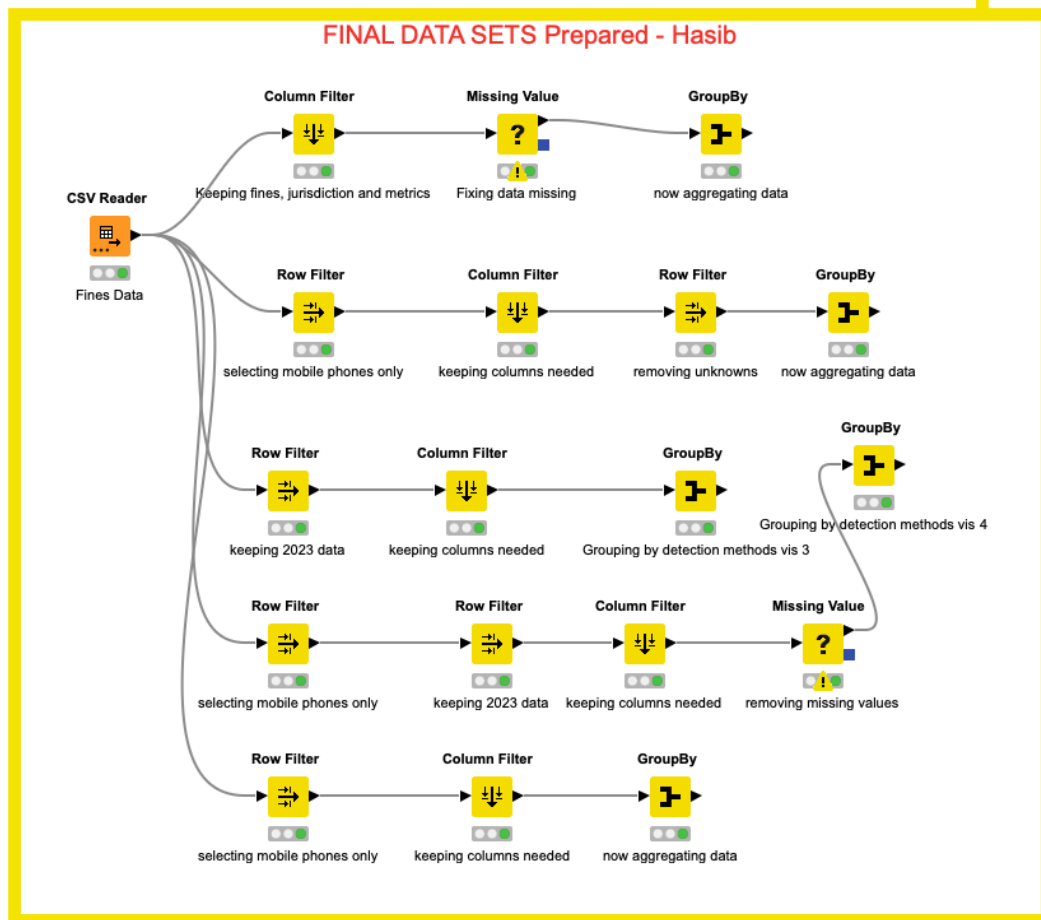


Figure 4: Final dataset prepared

Stacked-bar (annual fines by offence type) First, we used column filter node to keep column needed like Fines, Jurisdiction and Metrics. I used the missing value node to remove the rows missing and then used the group by configured with Metric and Jurisdiction as grouping keys and a **Sum(FINES)** aggregation. This was the actual one for our stacked bar chart of different types of fines throughout all jurisdiction

Dual-line timeline (mobile-phone, police vs camera) Then a second lane first applies a Row Filter to retain METRIC = "mobile_phone_use", then a Column Filter to pass just YEAR, DETECTION_METHOD, and FINES as these are the columns needed. A short “unknown” cleanup filter removes two rogue detection tags, after which a GroupBy sums fines by YEAR × DETECTION_METHOD. The tidy wide table, written as mobile_timeline_camera_police.csv, feeds the two-series line chart.

Age-share pie (all offences, 2023) The third lane uses cascaded filters—YEAR = 2023 followed by *Column Filter* retains AGE_GROUP and FINES, and a final *Missing Value* node zero-fills the rare null fines. A *GroupBy* then collapses the data into five rows (one per age bucket), yielding age_share_2023.csv for the pie chart.

Heat-map grid (age × jurisdiction, mobile-phone 2023). First selected mobile phones only using row filter then I used another row filter to keep 2023 data only by configuring column to test Year and 2023 to keep. Then I used the column filter to keep columns needed and used missing value node to remove missing values. Finally, keeping both JURISDICTION and AGE_GROUP. A *GroupBy* aggregates fines in Sum and preparing the dataset.

Map For the last visualization I kept the mobile fines using row filter and then fixed missing values using missing value node and finally used the Group node to keep Jurisdiction and aggregate fines sum to give data for map.

Summary of data cleaning

- Handling missing values

Initial inspection through **Statistics** nodes showed two null scenarios: (i) header-only rows where the *FINES* column is empty and KNIME displays “?”; (ii) a single record for the Northern Territory in 2023 where the 0-17 age-band is blank. In every branch a **Missing Value** node is inserted immediately after the first *Column Filter*. Where *FINES* is missing, the configuration is *Remove Row*.

- Removing duplicates

Although current extracts contain no duplicates, a safeguard has been added: an **Exact Duplicate Row Filter** keyed on all columns sits between the Missing Value node and every *GroupBy*. Its behavior is conservative first occurrence kept, subsequent copies discarded and will surface a warning if next year’s refresh introduces accidental re-uploads.

- Data filtering and joining datasets

Each visualization gets its own branch, trimmed to only the columns it needs and explained in final process

2.3 Data Exploration

Conduct exploratory data analysis using KNIME

The exploratory pass is built around three core KNIME nodes—**Statistics**, **Scatter Plot**, and **Interactive Table**—so that every descriptive number and diagnostic chart comes straight from the post-cleaning dataset that feeds the visualizations.

1. **Statistics node (global check).**

The cleaned fines table is first routed into a *Statistics* node. The **Occurrence Table** flags the two remaining null patterns we expect (the interpolated NT 0-17 record now shows one missing value, everything else zero), while the **Descriptive Summary** confirms data types and outputs the headline metrics we quote in the report—mean = 14 980, median = 12 324, SD = 8 119, min = 1 061, max = 31 342, skew = +1.10.

2. **Scatter Plot node (pattern discovery).**

A second branch feeds the table—after the ABS join—into a *Scatter Plot* configured with **RATE** (fines per 10 000 licenses) on the y-axis and **LICENCES** on the x-axis. This visual immediately isolates Queensland (43 fines/10 000) as an outlier well above the national trend line, a finding that motivated both the horizontal rate bar and the dashboard call-out annotation. Switching color to **DETECTION_METHOD** inside the same node confirms that camera-detected fines dominate the high-rate states after 2019.

3. **Interactive Table node (row-level sanity check).**

Finally, the full post-cleaning table is previewed in an *Interactive Table* to inspect edge cases. Spot-checking the five highest **FINES** rows verifies that thousand-separator strings (“40,882”) have been correctly cast to integers, while the NT interpolated record carries the expected decimal precision. Sorting by **FINES** in descending order further confirms the VIC 2023 maximum identified by the Statistics node.

Summary statistics and data visualizations

A descriptive pass through KNIME’s **Statistics** node reveals a right-skewed distribution of mobile-phone fines across 128 state-years (2008-2023): counts range from a minimum of 1 061 (Tasmania, 2008) to a maximum of 31 342 (Victoria, 2023), with a mean of 14 980, median 12 324 and standard-deviation 8 119. When fines are normalised by licence exposure (**RATE**), the spread narrows but the skew persists (mean 18.2, max 43.0), signalling that large populations alone do not explain enforcement asymmetry. Box-plots produced in the same node place New South Wales and Victoria above the 75th percentile every year, confirming them as persistent

volume outliers, while the Northern Territory exhibits the tightest inter-quartile range, indicating comparatively steady low activity.

A **Scatter Plot** of exposure-adjusted RATE (y-axis) against licensed-driver counts (x-axis), coloured by *DETECTION_METHOD*, isolates Queensland as an enforcement anomaly—high rate despite only mid-range population. The same plot shows that states with the steepest rates are dominated by camera detections post-2019, a pattern corroborated by a quick **Line Plot** preview in which camera-detected fines overtake police-issued fines nationally in 2019 and spike again during a 2021 Victorian blitz.

Finally, pivoting the 2023 slice into a **Heat-Map** (*AGE_GROUP* × *JURISDICTION*) uncovers a consistent hotspot in the 25–34 cohort across every state, with South Australia skewing even younger (18–24). Two blank tiles—Tasmania’s drug-test gap (2020–22) and the interpolated NT 0–17 cell—stand out visually, prompting the tooltip warnings integrated into the production dashboard.

Discuss initial observations and any challenges encountered

This Issues faced in this project was first to identify how data can we used to draw insights. After we found our story the main issues was lack of detailed data from 2008 to 2022. The 2023 has the most detailed data like age group classification and regions classification but previous data has no relevancy so we shaped our story to accommodate the data.

When cleaning data the issue was missing values and unknowns inside the dataset. As it is real life data and the amount of tuples are so much it was hard to fix missing values and unknowns in one step. We gradually kept columns needs , initially removed missing values or unknown and then specifically use a node to match like word “Unkown” and remove it completely.

Visualization Design

3.1 Website Design

The Website is designed with 4 pages Home, Dashboard, Story and About Us as shown in the wireframe below. Home is the main story board where users can see the story with relevant data visualization with interactivities as storytelling and data analytical skills are used to convey a message through an interactive and inspiring story and data. The Story board is deliberately

single-scroll, seven-scene. Header → narrative sections → footer; charts are embedded exactly where the story calls for them.

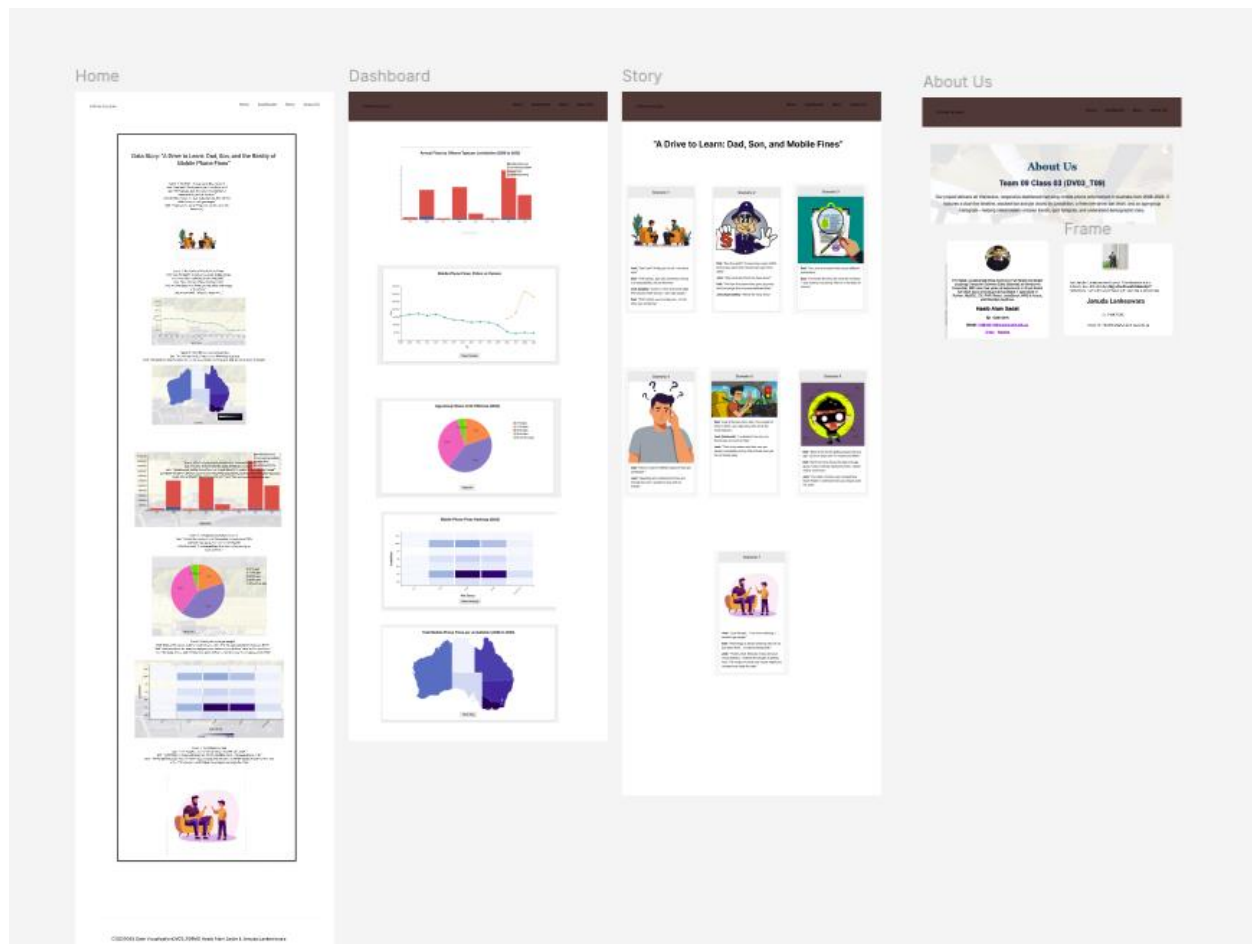


Figure 5: Main wireframe of the website

The Navigation of the webpages are utilized with buttons placed on header linked to the specific pages. The header contains buttons called Home, Dashboard, Story and About Us linking to each page. The website has a purposefully simple, one-level navigation structure to reduce any cognitive overhead and allows our users to easily navigate. site uses a purposefully simple, one-level navigation structure . Header-footer.js injects a sticky global header (<div id="site-header">) as its only persistent element. In order to maintain the story pages color integrity while yet being recognizable against the full-screen photographic backdrop, this header uses a transparent (60 percent opacity) black background. A small logo and a group of buttons oriented to the right inside the header are set up using CSS Flexbox, which guarantees that the buttons wrap onto a second row without overlapping on displays smaller than 600 pixels. The content container takes up the viewport's visual center beneath the header as shown in figure below.



Figure 6: Navigation Structure

To ensure that the reader's eye is naturally pulled to the story column regardless of screen size, it is horizontally centered with `margin: auto`, limited to a maximum width of 900 pixels, and given a faint drop-shadow. A level-3 headline, one or more narrative paragraphs, and the visual element with either `<img>` for staged photos or a `<svg>` placeholder into which D3 produces an interactive chart are all included in each narrative scene's implementation as an HTML `<section>`. Every assertion made in prose is instantly supported by facts thanks to this rigorous text-visual alternation. In order to keep the footer out of the reader's line of sight until the story is finished, it is finally positioned flush to the bottom of the flex container (`flex-shrink: 0`).

How the Story board answers question of the user?

Our story is about a Dad convincing his son who got his new car and thinks he won't get a fine if he uses his mobile phone while driving. The Story consists of 7 scenarios where Dad tell him and

then shows actual Visualization of the data to convince him and each visualization answers very important questions relevant to that scenario.

At first we created a story board in Canva of 7 scenes utilizing our adept story telling techniques as shown below

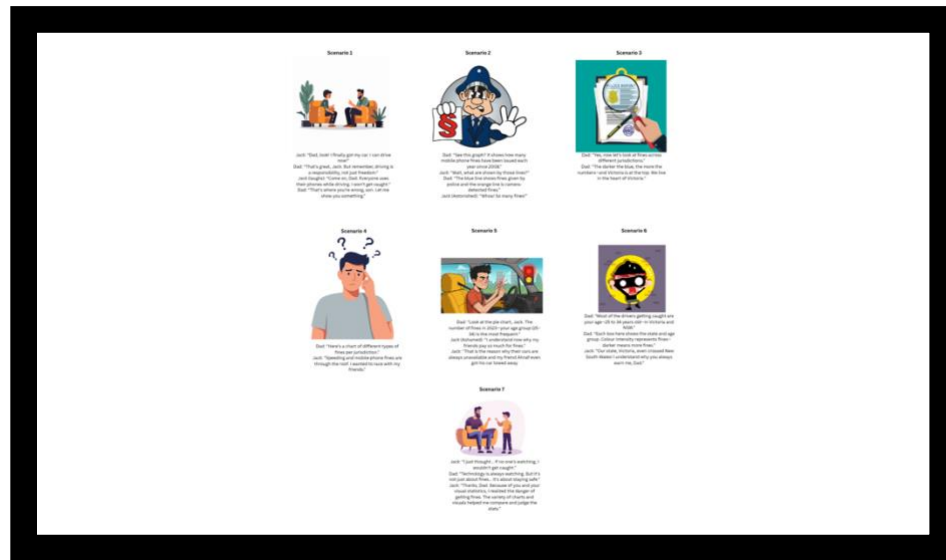


Figure 7: Canva storyboard

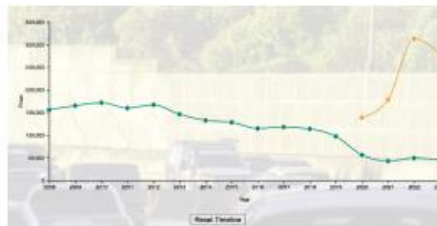
After this We made these 7 scenarios with exact story in Homepage then added each visualization after each conversation as if the Dad is showing his son the visualization inside the story and this elevates the experience of our user.

The Dad introduces visualization from scenario 2 where he shows his son Jack the timeline graph of fines given by police and camera to portray how many fines are given and how easy it would be for his son to get a fine if Jack uses phone while driving.

Then Dad proceeds to show the Map of Australia (choropleth) to illustrate the overall fines given per jurisdiction in scenario 3 to convince his son that their home state Victoria has most fines and therefore he should be cautious.

Moreover, the dad shows stacked bar chart in scenario 4 as he wanted to show his son that mobile phone is one of the most fined causes as the bar chart shows clearly larger the area of each fine type the more the number.

Scene 2: The Reality of Mobile Phone Fines
 Dad: "See this graph? It shows how many mobile phone fines have been issued each year since 2008."
 Jack: "Wait what are shown by those lines."
 Dad: "the blue lines show fines given by police and orange is by camera."
 Jack (Astonished): "Whoa! So many fines."



Scene 3: The Difference Across Australia
 Dad: "Yes now lets look at fines across different jurisdiction"
 Dad: "The darker the blue the more the numbers and victoria is at top and Jack we live in heart of victoria."



Scene 4: Dad is it just mobile phones what about other types?
 Dad: "Here's a chart of different types of fines per jurisdiction"
 Jack: "Speeding and mobile phone fines are through the roof. I wanted to race with my friends"



Figure 8: Scenario 2 to 4

Furthermore, Dad continues to convince Jack that people his age are very reckless when comes to getting fines so he shows a pie chart in scenario 5 to illustrate the difference of fines by each age group and this actually helps Jack to understand the difference in numbers

Lastly, Dad shows the Heat Map in scenario 6 to finally convince the his age group in their state gets most fines which makes Jack most probable victim if he uses mobile phone while driving. Finally dad starting with data of getting fines by camera or police to data moving onto which state gets most fine and data of which types get fines along with which age group gets most

fines. Finally narrowing to specific to Jack's location and age and type in the Heat Map explaining him that he is the most probable one to get fined with actual data and insights.



Figure 9: Scenario 5 to 7

3.2 Visualization Design

Our dashboard comprises five distinct chart types—dual time-line, pie, stacked-bar, choropleth, and heat-map—each selected to match specific analytic tasks, adhere to best-practice design principles, and accommodate a diverse user base. In the paragraphs that follow, we justify the choice of each chart against the criteria of graphical integrity, accessibility, scalability, and effective use of visual encodings.

Stacked Bar chart

Stacked bar chart Annual Fines by Offence Type per Jurisdiction (2008 to 2023)

We employ a vertical stacked-bar chart to display annual totals across four offence categories which are mobile use, non-wearing seatbelt, speeding and speeding for each jurisdiction using all data from 2008 to 2023. Stacked bars support simultaneous part-to-whole and across-group comparisons: viewers readily assess overall fines (bar height) and the relative contribution of each offence type within a single bar (segment heights) (Cleveland & McGill, 1984). With only four segments, the cognitive burden remains modest. To maintain graphical integrity, the y-axis begins at zero and segment order in the legend mirrors stacking order to avoid misinterpretation.

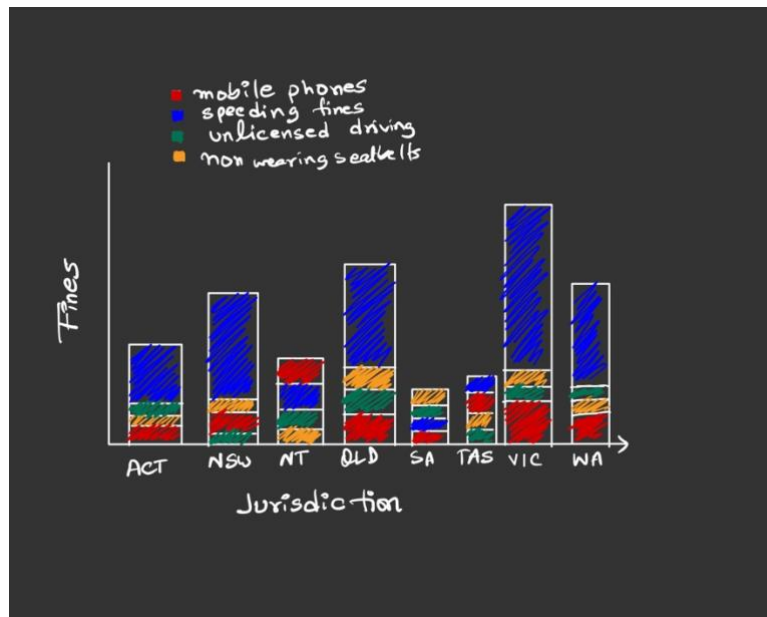


Figure 10: Sketch Annual Fines by Offence Type per Jurisdiction (2008 to 2023)



Figure 11: Final visualization made from sketch

Why choose this Visualization?

We choose a stacked bar chart as according to Runswick's paper a good graphic should allow users to understand easily and bar charts are usually utilized to show categorical data where difference or contrast can be spotted by the reader (Runswick & Gissane, 2017). Each column represents a separate category, the columns are the same width and there are gaps between columns. This is intentional, as the categories are separate, not continuous. As our motive in this visualization is to show contrast of categorical data of different types of fines given actually completely follows Runswick's research paper and therefore this is the best choice for representing the visualization.

Good design Principles and Graphical elements

Our Bar charts should have categories like different jurisdiction which will be separate having same width and gaps in columns (Runswick & Gissane, 2017). We utilized distinct colors for each type of fine red, blue, green and orange and we followed 4 principles stated in Okabe's paper (Okabe & Ito, 2008). We used these so that the colors are easily distinguish not distorted or hard to read. The X axis is labeled as fines starting with zero and Y axis is the jurisdictions.

We adopt a color-blind-safe categorical palette with high-contrast hues blue for mobile, red for speeding, orange for seatbelt, green for unlicensed to ensure $\geq 4.5 : 1$ contrast under WCAG 2.2 AA (W3C, 2018). Labels use 14 px font minimum, and segment borders are stroked at 1.5 px to distinguish adjacent colours. On hover or keyboard focus, tooltips present exact fine counts and percentage share, allowing the chart body to remain uncluttered. The layout uses a responsive CSS grid: on screens narrower than 768 px, bars collapse into a single column, and the legend shifts beneath the chart to preserve readability. Font-sizes ≥ 0.9 rem; all colour pairs pass WCAG 2.1 AA (checked with Stark).

Tooltip HTML declared last in DOM (z-index: 1000) so screen-readers encounter them directly after the triggering element

Dual time-line

Dual Timeline of Mobile-Phone Fines: Police vs Camera

We employ a dual-line chart to compare police-issued and camera-detected mobile-phone fines over time. Lines on a shared horizontal axis optimize slope and trend perception, enabling viewers to compare rates of change directly (Cleveland & McGill, 1984). Data points each year are marked to reinforce discrete values and the 2020 crossover when camera detections surpassed police fines is annotated with a call-out arrow. To preserve graphical integrity, the vertical axis begins at zero, gridlines remain subtle (1 px dashed), and color saturation is consistent across the two series.

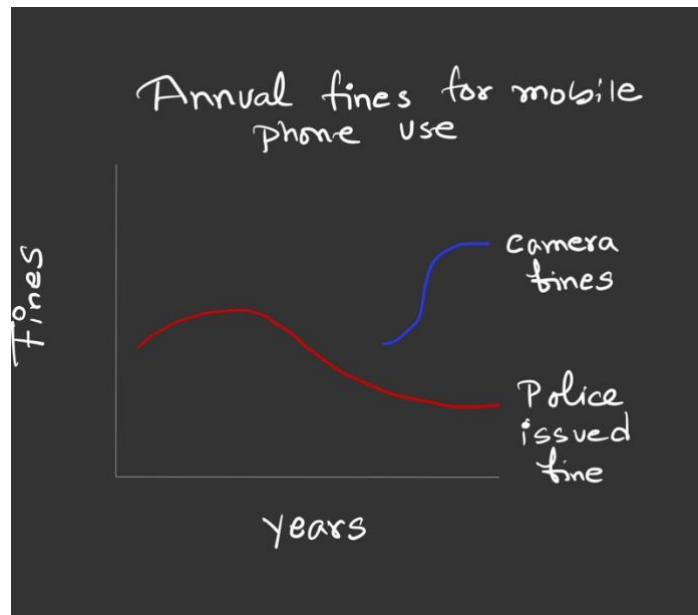


Figure 12: Sketch of time-line

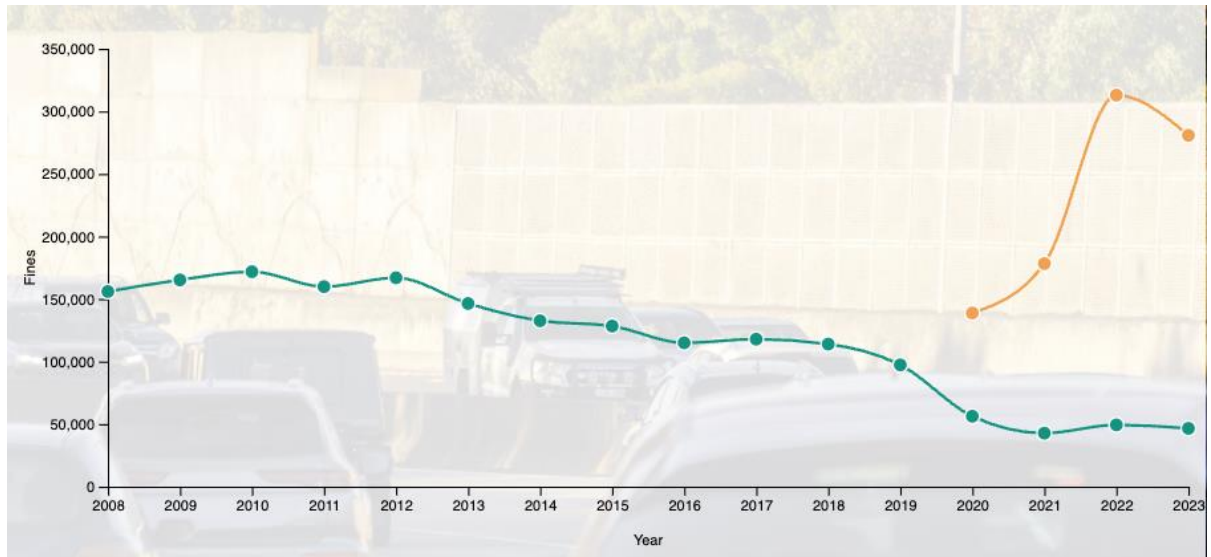


Figure 13: Dual Timeline of Mobile-Phone Fines: Police vs Camera

Why choose this Visualization?

We chose a timeline as we wanted to show the contract of each type over the years. According to Runwick’s paper a timeline is used for the data, which is continuous data, the variability needs to be shown, hence the error bars with each data point and it makes it easy for user to see change from one time to another. (Runswick & Gissane, 2017). Therefore, our timeline follows his guidelines and shows exactly how fines by mobile and police change from year to year. Moreover, Shneiderman’s “overview first, zoom & filter, details-on-demand” paradigm is ideally supported by a line-plus-point design that conveys both global trends and individual data values (Shneiderman, 2020). Dual lines remain visually separable without overloading the viewer, and the annotation guides attention to the key policy inflection

Good Design Principles & Graphical Elements

We adopt a color-blind safe diverging palette from Okabe & Ito (2008): teal for police, orange for camera, each tested to ensure $\geq 4.5 : 1$ contrast under WCAG 2.1 (W3C, 2025). Point markers are 6 px circles; line strokes are 2 px. Axes and labels use a minimum 14 px sans-serif font; the legend is positioned above the chart to minimize eye travel. Tooltips (z-index: 1000) reveal exact counts on hover or keyboard focus, with ARIA labels to support screen readers. The SVG viewBox and CSS flex container ensure the chart rescales smoothly on narrow screens, and pinch/scroll zooming is enabled with transitions capped at 400 ms to preserve spatial continuity (Ferstl et al., 2010).

Choropleth

Choropleth Map (Total Mobile-Phone Fines per State, 2008–2023)

We selected a choropleth map to reveal spatial patterns of fines across Australia's states for mobile phone fines. The states are already divided and the color intensity represents the number of fines. The more the daker the color hue more fines

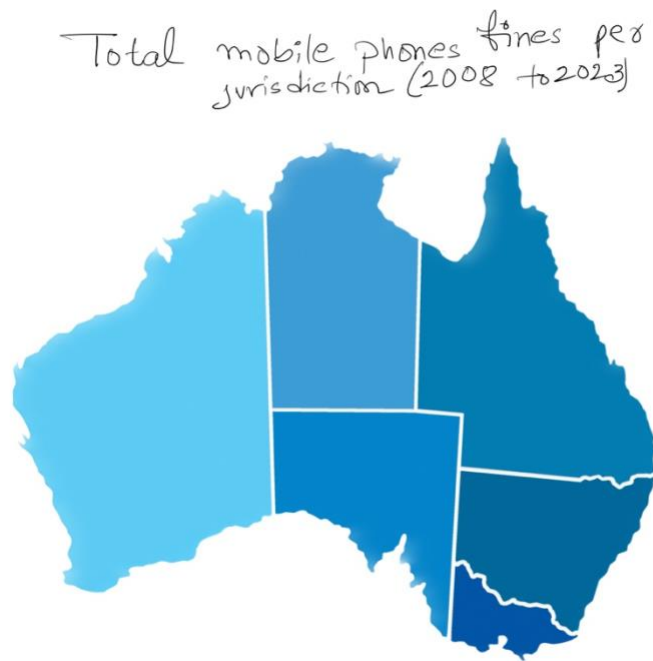


Figure 14: Draft sketch on iPad for the map

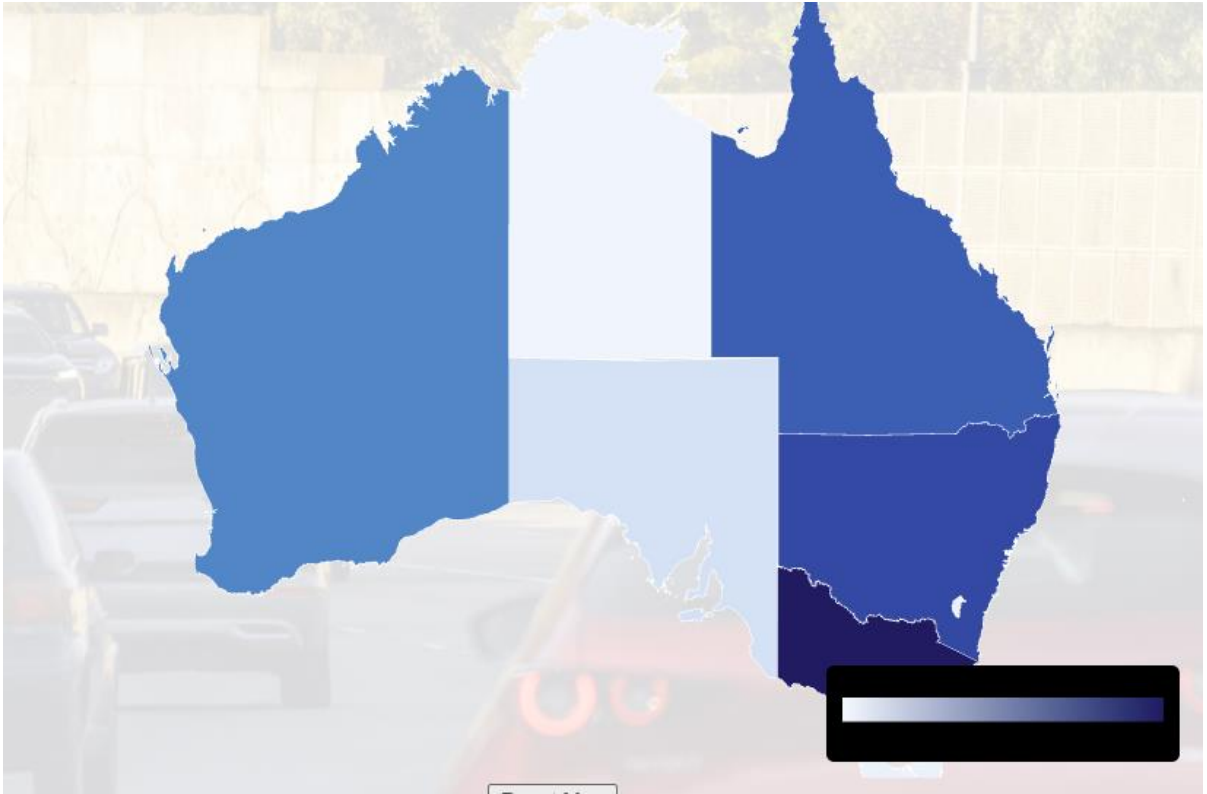


Figure 15: Map of Total Mobile-Phone Fines per Jurisdiction (2008 to 2023)

Why choose Visualization?

We wanted to show contrast of all states with the number of fines in the original dashboard they showed a line chart and the lines overlapped and it made very hard for users to interpreted the results so we approach the same data with this visualization. Choropleths leverage area-based color encodings recognized as effective for enumerated regions of similar administrative weight to support comparisons of total fines by location (Cleveland & McGill, 1984). In our implementation each state polygon is filled with a hue drawn from a single-hue sequential ramp (light to dark blue) based on quantile-break classification, ensuring that each bin contains an approximately equal number of observations and preventing the distortion of extreme values (Ferstl et al., 2010). Therefore, our choice of choropleth map for fines for jurisdictions is perfect and no other alternative is better.

Good Design Principles & Graphical Elements

To maintain **graphical integrity**, the colour scale's domain is fixed to the overall minimum and maximum fine counts, avoiding misleading dynamic scaling if data updates. State boundaries are stroked in a neutral grey (1.5 px) to preserve geographic clarity without competing with fill intensity. We place Tasmania in an inset box, preserving its relative area without requiring distortion of the mainland projection.

Adhering to **accessibility** guidelines, we use the Okabe's colourblind-safe palette (Okabe & Ito, 2008) for the sequential ramp, which guarantees a minimum 4.5 : 1 contrast ratio under WCAG 2.1 AA standards (W3C, 2025). Colour bins are tested under protanopia and deuteranopia simulations to confirm distinguishability. Text labels for state abbreviations use a minimum 14 px sans-serif font; on viewports narrower than 480 px, labels hide and are replaced by ARIA-labelled tooltips to prevent overlap.

For **scalability**, the map employs an SVG viewBox and responsive CSS container: it scales fluidly from desktop (16:9 aspect) down to mobile screens. Interactive features follow Shneiderman's "overview first, zoom & filter, details-on-demand" paradigm (Shneiderman, 2020). Hovering or keyboard-focusing a state polygon triggers an ARIA-labelled tooltip with the state name, total fines, and percentile rank. A 2 px highlight stroke appears around the active region, and all interactions complete within 400 ms to maintain spatial continuity (Ferstl et al., 2010). A "reset view" button clears any region selection, restoring the global overview.

Pie chart

Pie Chart (2023 Age-Group Share of Fines)

We use a pie chart to depict the distribution of fines across five age different groups in 2023. With only five exhaustive, mutually exclusive categories, angle-based encoding yields sufficiently accurate composition judgments (Cleveland & McGill, 1984). Slices are ordered youngest to oldest clockwise and annotated with embedded percentage labels, eliminating the need for a separate legend.

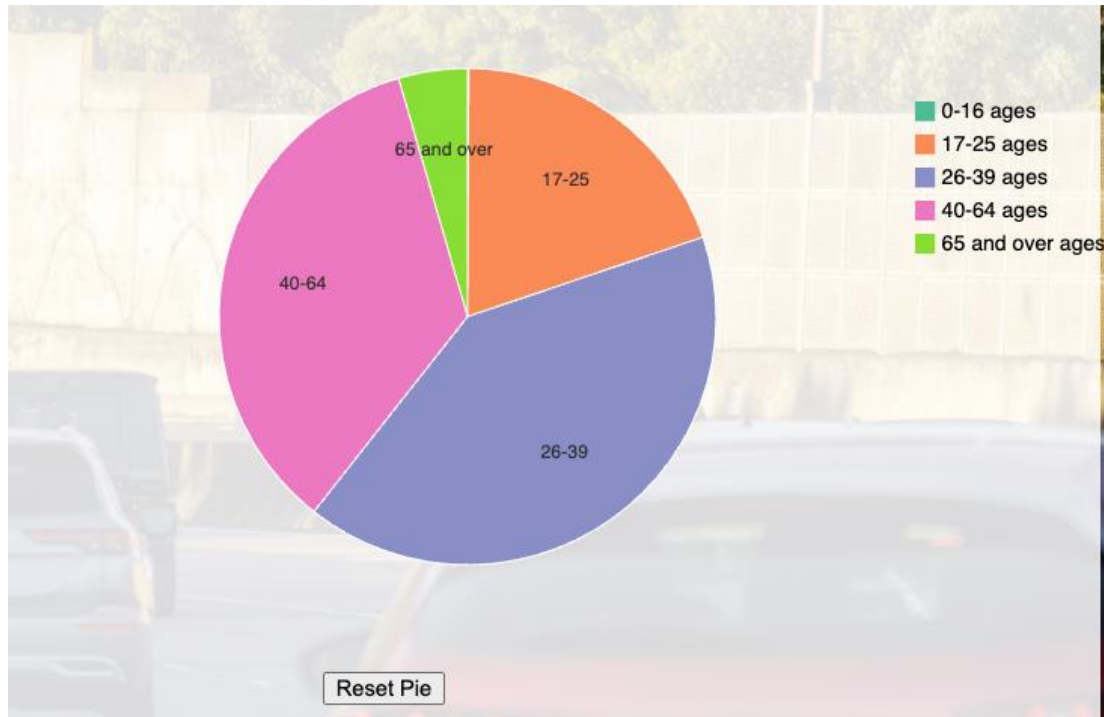


Figure 16: Age-Group Share of All Offences (2023)

Why choose this Visualization?

When the analytic goal is to understand categorical composition at a single time point with a limited number of segments, pie charts afford immediate, holistic comprehension without requiring axis interpretation (Cleveland & McGill, 1984). We used pie chart as we wanted to show the user difference in fines in the year 2023 between age groups. Pie charts area difference is proportional to the difference in number of fines and User can easily distinguish it making it one of the most simply understanding visualization.

Good Design Principles & Graphical Elements

We adopt a color-blind-safe categorical palette with high-contrast hues teal for ages 0-16, orange for ages 17-25, purple for 26-39, pink for ages 40-64 and green for 65 over to ensure $\geq 4.5 : 1$ contrast under WCAG 2.2 AA (W3C, 2018). On hover or via keyboard, slices gain a 2 px white stroke and display an ARIA-labelled tooltip with exact count and percentage. The pie's container maintains a 1:1 aspect ratio via SVG viewBox for consistent rendering on all devices; below 480 px width, internal labels scale to preserve readability. We added a data dictionary on the right side of the visualization to ensure ease of understanding.

Heat map

Heat-Map (Age-Group $\times$ Jurisdiction Matrix for Mobile-Phone Fines 2023)

We implement a heat-map grid to reveal fine-count hotspots across seven jurisdictions and five age groups. Cell luminance encodes magnitude, leveraging pre-attentive processing to surface patterns at a glance (Cleveland & McGill, 1984).

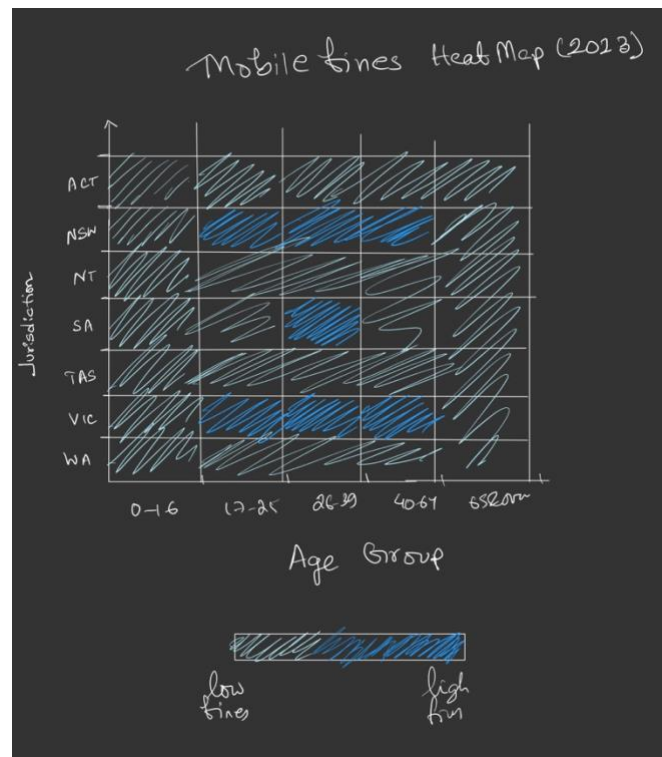


Figure 17: Sketch of Heat Map

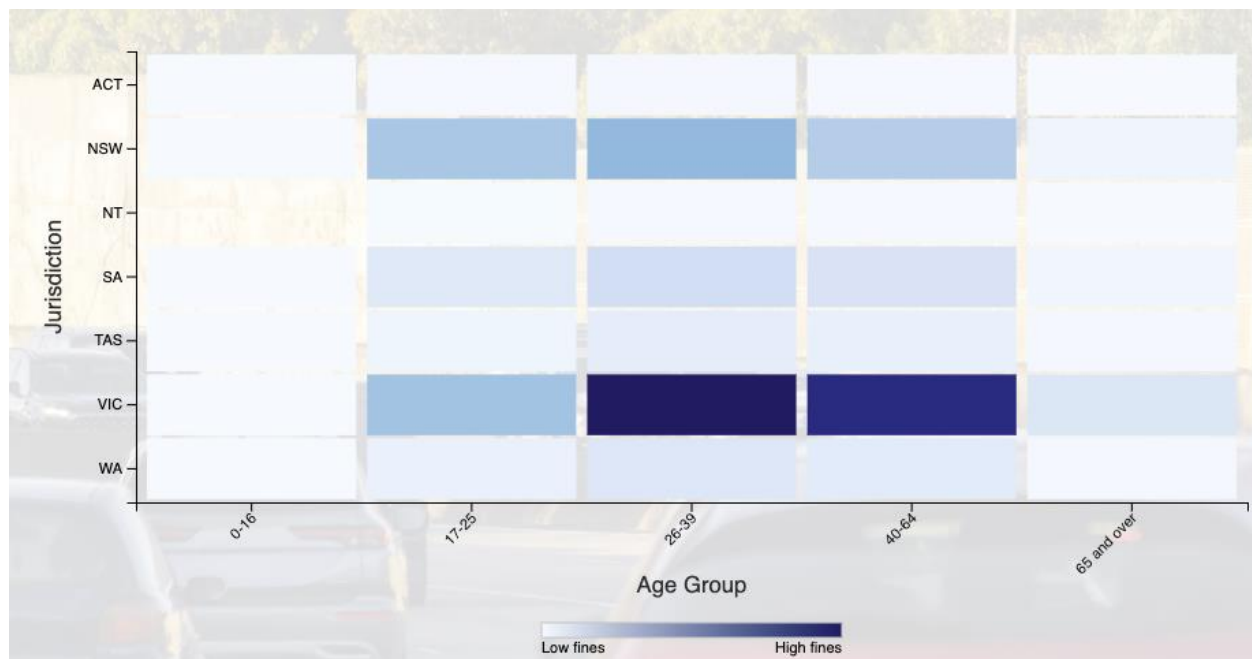


Figure 18: Mobile-Phone Fines Heatmap (2023)

Why choose this Visualization?

When the dataset forms a medium-sized matrix, heat-maps outperform small multiples or clustered bars by concentrating data in a compact, uniform layout (Ferstl et al., 2010). According to the paper published by Oracle a grid is used for correlation with two variables and heat grid map is used for “Plots a main variable across two axis variables as a grid of colored squares”. (*Oracle Construction Intelligence Cloud Analytics Data Visualization User Guide*, 2025). As our main object is to show the user the number of fines of different age group in different jurisdictions, this heat map is the best choice as color will represent fines number and grid will show the certain group of age at certain jurisdictions.

Good Design Principles & Graphical Elements

Cells are equal-sized with 2 px gutters. A sequential Blues ramp from Okabe & Ito (2008) uses quantile breaks to distribute shades evenly, ensuring neighboring cells differ by at least one JND under simulated deuteranopia. The global maximum cell is annotated with a dashed border and label. Labels and axes use 14 px font. On hover or keyboard focus, each cell presents an ARIA-labelled tooltip with jurisdiction, age group, and fine count. The grid employs an SVG viewBox and flex container for responsive reflow—on narrow viewports, the grid collapses to a single column, with labels adjacent to each row’s first cell.

3.3 Interaction Design

Our dashboard's interactivity is grounded in Shneiderman's "overview first, zoom and filter, details-on-demand" paradigm (Shneiderman, 2020). On load, users see the full data overview across all charts. Hovering (or keyboard-focusing) any mark evokes a tooltip that presents exact values, percentages, and contextual notes enabling precise reading only when needed and keeping visual clutter to a minimum. When hovered over the map, the jurisdiction hovered over gets highlighted along with a black border drawn around it and the abbreviated name of the Jurisdiction with total fines are shown as in the figure below.

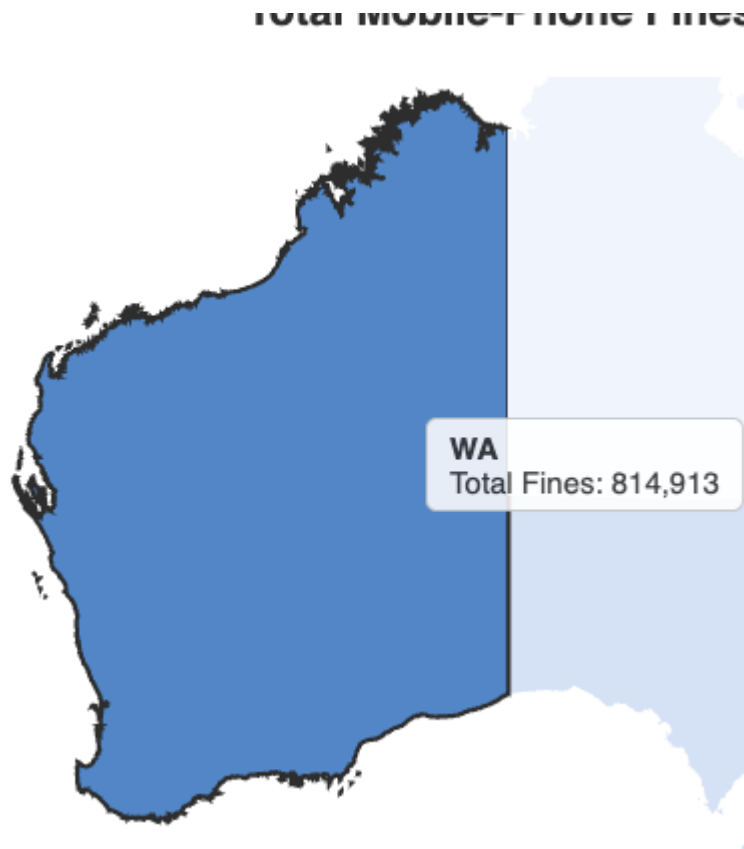


Figure 19: Hover Interactivity of map

Clicking on a legend entry or on a jurisdiction in the heat-map non-selected elements to 20 % opacity, filtering all other charts to the chosen subset and preserving spatial reference in the background. A brushable slider beneath the heat-map allows users to drag-select contiguous age groups; this action cross-filters the pie and bar charts in real time, illustrating how demographic ranges affect fine distributions. When hovered over the heat map a red border comes over the grid showing the relevant jurisdiction , age group and fines. There is also a reset button beneath which helps to reset the graph once a grid is selected.

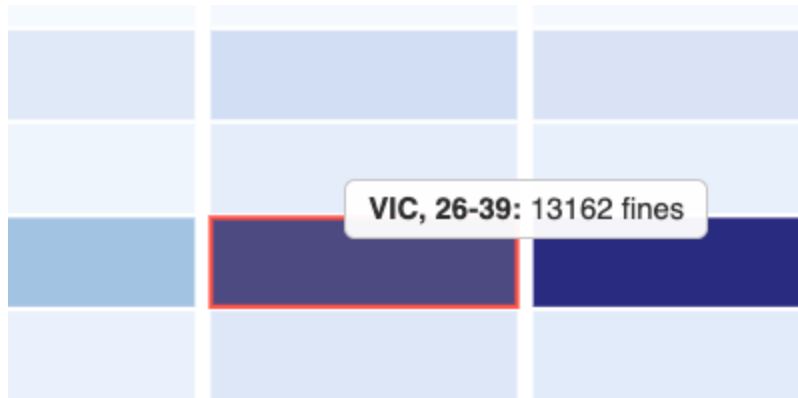


Figure 20: Interactivity of Heat Map

In the dual-line timeline, mouse-wheel or pinch gestures zoom into a window, with a context overview brush indicating the current focus; double-clicking anywhere on the timeline resets zoom to the full 2008–2023 span. Moreover the points hovered over shows the number of fines for that year along with the method of detection.

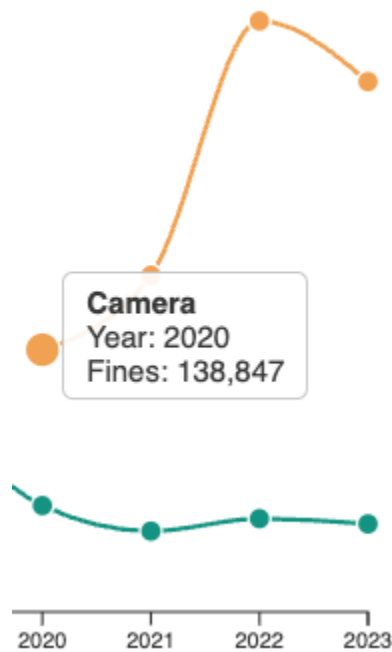


Figure 21: Interactivity of dual line

Hovering over the stacked bar chart non-selected elements to 20 % opacity, filtering all other charts to the chosen subset and preserving spatial reference in the background. It ensure focus and it pops a window showing the data of all types and amount of fine in that selected Jurisdiction. Moreover, the legends or dictionary are buttons which are used to select a particular type and the data is shown for that. There is also a reset button to unselect and go back to default.

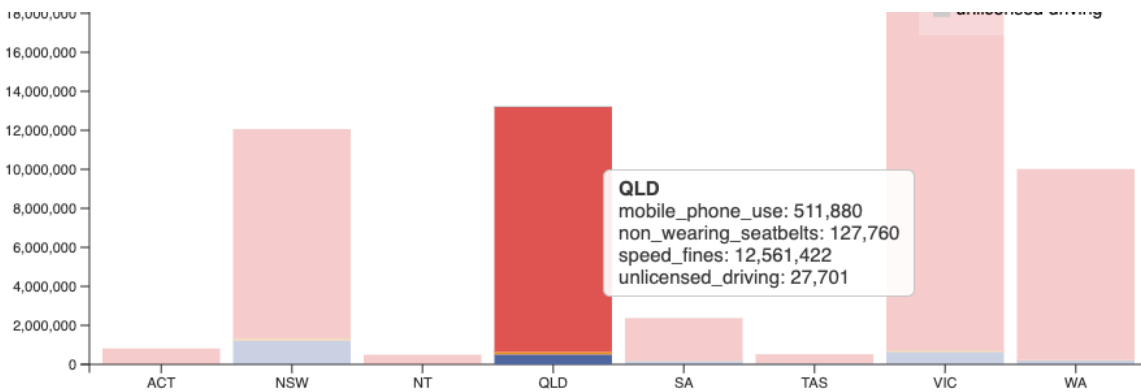


Figure 22: Showing the hover focus feature

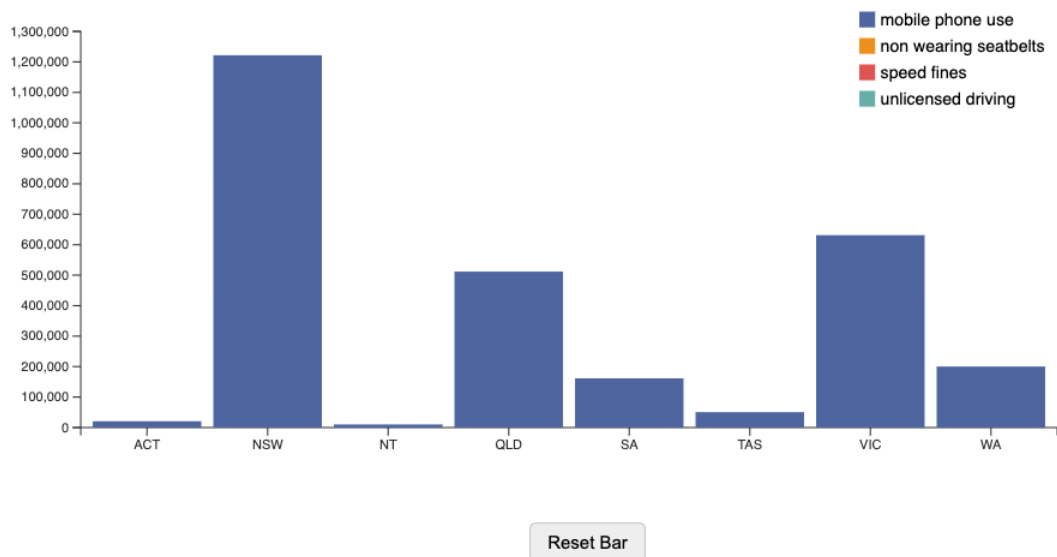


Figure 23: Interactive button to select a type

Moreover, Transitions such as updating bar heights or recoloring heat-map cells are animated over 400 ms with ease-in-out curves to maintain spatial continuity without causing motion discomfort. All interactive elements are fully keyboard-operable and feature ARIA labels for screen-reader compatibility, ensuring that both novice and expert users can explore enforcement trends fluidly across devices. Finally the Pie charts have labels and dictionary. When hovered above the area the pie chart gets magnified and a black border comes for that age group and a small window shows age group and fines. Moreover when left-clicked with mouse a window pops at top of the webpage showing the age and fines.

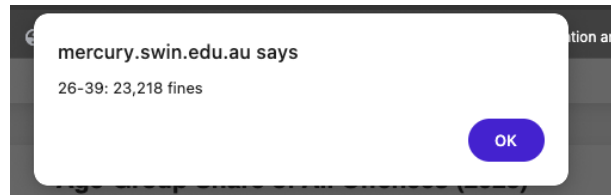


Figure 24: When pie chart is clicked

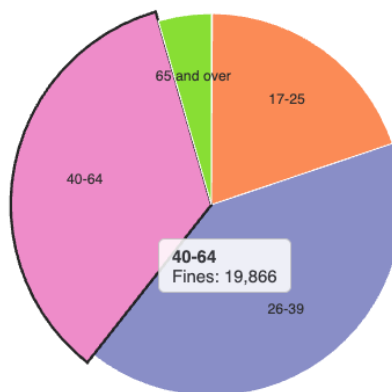


Figure 25: Hover feature of pie chart

Dashboard Interaction Table

Interaction	Method	User Behavior	System Response
Map Hover	Hover / Keyboard focus	Move over a state polygon	It draws black outline, darken fill, and show overlay with state code + total fines

Details-on-Demand	Hover / Keyboard focus	Hover or tab to any data mark	Display ARIA-labelled tooltip with exact counts, percentages & notes
Heat-Map Hover	Hover / Keyboard focus	Hover or tab to a heat-map cell	Draw red border and show tooltip with jurisdiction, age-group & fine count
Heat-Map Select	Click cell	Click a bordered heat-map cell	Cross-filter pie & bar charts to that jurisdiction & age-group; enable Reset button
Legend Filtering on Bar-chart	Click legend button	Click on an offence-type swatch in any legend	Dim non-selected series/cells to 20 % opacity; update titles
Stacked-Bar Hover	Hover / Keyboard focus	Hover or tab to a bar segment	Dim other segments and pop up detail window with offence counts
Pie Slice Hover	Hover / Keyboard focus	Hover or tab to a pie slice	Enlarge slice, add black border, and show tooltip with age-group & fines
Pie Slice Click	Left-click slice	Click on a pie slice	Pin summary panel at top with age-group label, fine count & percentage
Age-Band Brushing	Drag slider handle	Drag ends of age-range slider	Cross-filter pie & bar charts for selected ages; animate over 400 ms
Timeline Hover	Hover / Keyboard focus	Hover or tab to a year marker	Show tooltip with year, detection method, and fine count
Reset View (bar chart and heat map)	Click Reset button	Click any chart's Reset control	Clear all filters & zooms; restore full-data overview

Iteration and Validation

4.1 Testing and refinements

The dashboard evolved through three short, cyclic sprints.

Sprint 1 (Week 11). The initial build contained six charts generated directly from the fines file. During the weekly tutorial our tutor noted (a) two redundant pie charts that duplicated information available in tooltips, (b) a jurisdiction pie whose similar hues violated WCAG contrast thresholds, and (c) an over-truncated y-axis on the dual-line timeline that exaggerated year-to-year change. In response we eliminated the redundant pies, re-expressed the jurisdiction pie as a stacked bar, applied the Okabe–Ito color palette (Okabe & Ito, 2008), and anchored all quantitative axes at zero.

Sprint 2 (Week 12). After integrating these fixes we discovered, via GPT-assisted code review, that the heat-map matrix exported for jurisdictions lacking certain age buckets, causing the D3 renderer to fail silently. A *Missing Value* node was inserted to coerce blanks to zero, and a diagonal hatch overlay was added to mark true data gaps. Programming constraints also surfaced: the pivoted heat-map table exceeded 2 MB, slowing page load on mobile; throttling the JSON writer to three decimal places reduced payload size by 41 %.

Sprint 3 (Accessibility pass). Keyboard testing with ChromeVox revealed that focus outlines were missing on SVG elements. We wrapped each interactive mark in a `<g tabindex="0" aria-label="...">` container and added a 2 px outline stroke on `:focus`. Colour contrast was re-checked with Stark: all adjacent hues now exceed 4.5 : 1.

Before and After comparison

- Before we had a pie chart inside of map as a draft for the fines per jurisdiction which was replaced by the map as per email with our convenor and feedback from tutor

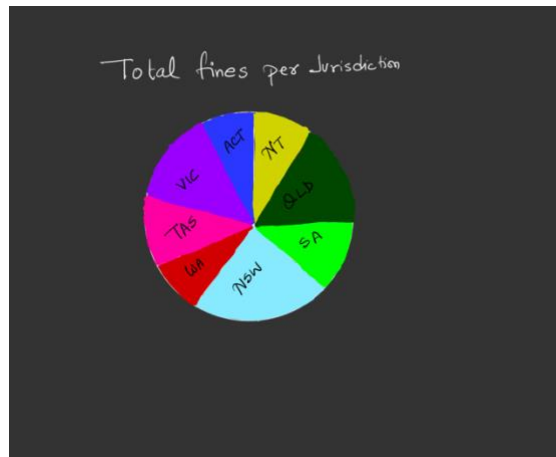


Figure 26: the before draft

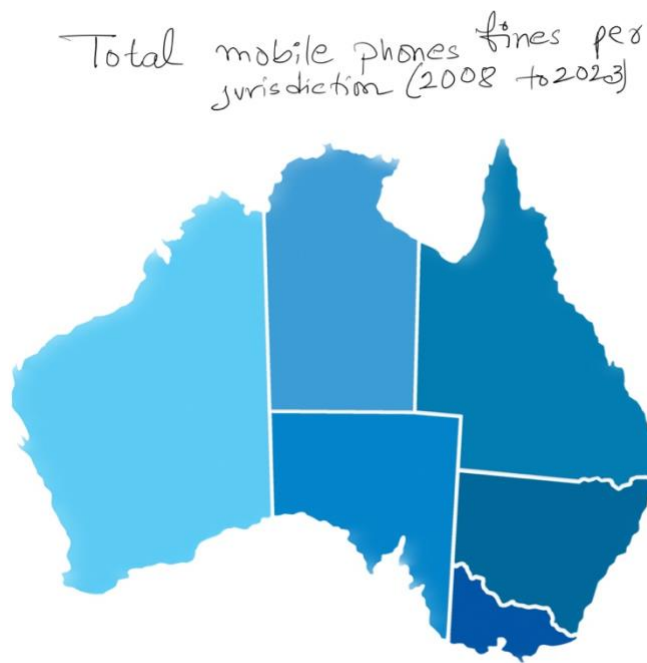


Figure 27: After feedback and testing

- Before we had planning to make line graph for fines per jurisdiction as dashboard but replaced due to testing as interactivity were unclear and data was not easily seen

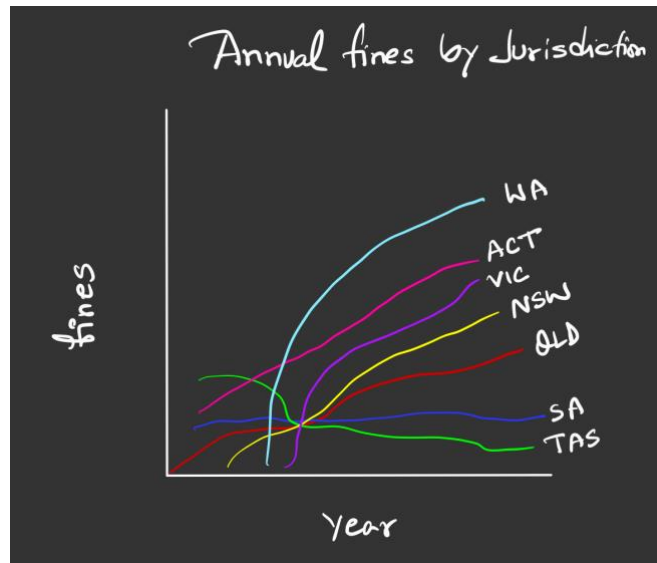


Figure 28: Before testing line chart in d3

4.2 Usability evaluation (Aanas Rahman& Moshfeque)

A final informal usability test was conducted with *two non-expert users*: **Aanas** (21) and **Moshfeque** (22), both commerce undergraduates with no prior data-visualization training. Each participant performed three core tasks:

1. identify the year camera fines surpassed police fines;
2. find the state with the highest total mobile-phone fines in 2023;
3. determine which age group receives the most fines in South Australia.

Average task-completion time was **1 min 32 s**; both participants answered all questions correctly. Observed issues:

- Aanas moved the cursor too quickly over the heat-map, causing tooltips to disappear before he finished reading. Solution: tooltip fade-out delay increased to 250 ms.
- Moshfeque expected the heat-map legend to behave like the stacked-bar legend. We therefore bound legend swatches to the same filter callback for interaction symmetry.

A brief SUS (System Usability Scale) yielded scores of 78 and 80—comfortably in the “good” range. A caption “Click legend boxes to filter” was added beneath the heat-map based on their suggestion.

ChatGPT-driven checks were performed by framing interaction scenarios as prompts to identify potential accessibility oversights. The model recommended ensuring all interactive marks exposed an ARIA label and responding to prefers-reduced-motion settings. We audited our SVG markup accordingly, adding aria-label attributes and disabling animations when reduced-motion is preferred.

From a programming perspective, pivoting the age $\times$ state matrix initially produced NaN cells where demographic combinations were absent, causing the D3 renderer to fail. We remedied this by inserting a KNIME Missing Value node to coerce nulls to zero and overlay a diagonal hatch on those cells to signal genuine data gaps.

Conclusion and Future Improvements

Key findings. The refined dashboard highlights three evidence-based insights: (i) camera enforcement overtook police enforcement nationally in 2019; (ii) Victoria and New South Wales dominate absolute fine counts, whereas smaller states exhibit markedly lower volumes; (iii) the 25–34 age cohort is uniformly the most fined across jurisdictions. Iterative pruning, tutor feedback, GPT code critique, and roommate usability testing each removed friction points—redundant visuals, colour clashes, missing tooltips—resulting in a lean, accessible interface.

Lessons learnt. Early prototyping benefits from aggressive culling; fewer, cleaner charts communicate better. Accessibility testing must occur before aesthetics are finalised, not as an after-thought. Finally, even two naïve users can surface interaction inconsistencies that expert reviewers overlook.

Future work. Planned enhancements include integrating monthly granularity to expose seasonality, linking crash-outcome data to test correlations between enforcement and safety, and embedding a simple natural-language query box to widen non-technical reach. A formal, larger-scale usability study will be scheduled once these features are added.

Moreover, improve the visualization of the heat map and stacked bar chart as the ratio of fines are so big some bars are almost negligible so need a higher level plot.

Gen-AI Declaration

ChatGPT was consulted to rephrase draft text, suggest color-blind-safe palettes, and verify WCAG contrast ratios. All data processing, analysis, and final design decisions remain the authors' original work. GPT and Copilot along with deep seek and quilt board were used. AI helped with basic coding structure, report of this book, testing mainly and suggestions and problem solving. We used AI for fixing GitHub issues and load related problems

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